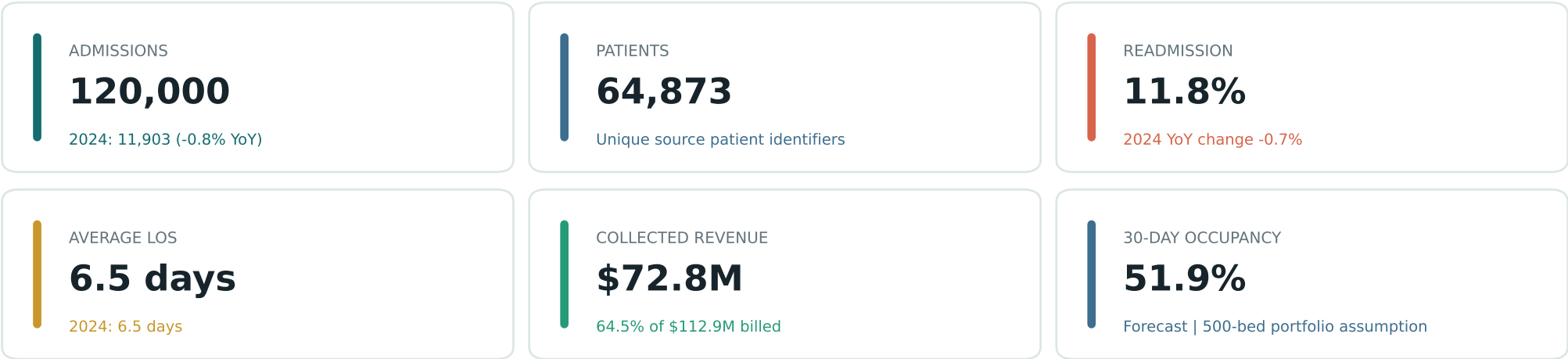


Hospital Operations Intelligence

Executive decision brief | Observed KPIs lead; assumptions are explicitly labelled

Period: 01 Jan 2015 - 30 Dec 2024 | Generated: 31 Jul 2026 06:58 UTC



DECISION PRIORITIES

- 1

ICU readmission review

4,950 of 20,833 ICU encounters were readmitted within 30 days (23.8%).

Clinical Quality Lead | 30 days
- 2

Revenue-cycle leakage

Collected 64.5% of billed value. Approval is 91.4%, showing that approval and collection are different controls.

Revenue Cycle Manager | 30 days
- 3

Operational data readiness

Waiting time is simulated and admission-level revenue is surrogate-linked. Replace these fields before operational deployment.

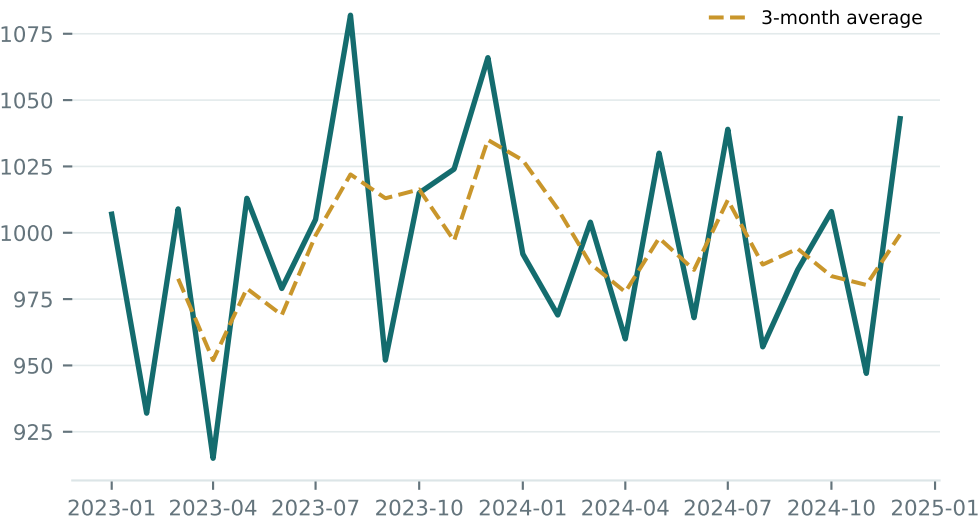
Operations Data Owner | 60-90 days

Patient Flow and Outcomes

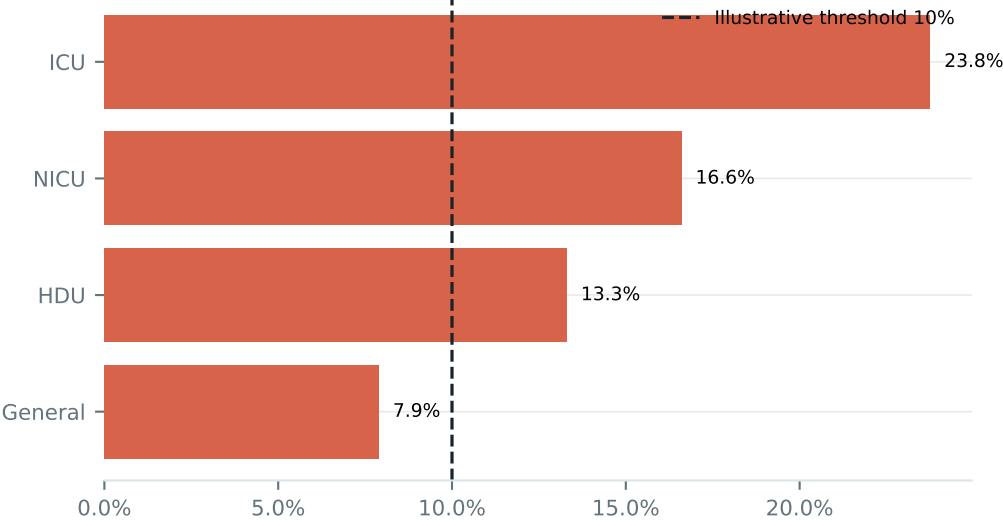
Observed admissions, length of stay, and readmission outcomes

Period: 01 Jan 2015 - 30 Dec 2024 | Generated: 31 Jul 2026 06:58 UTC

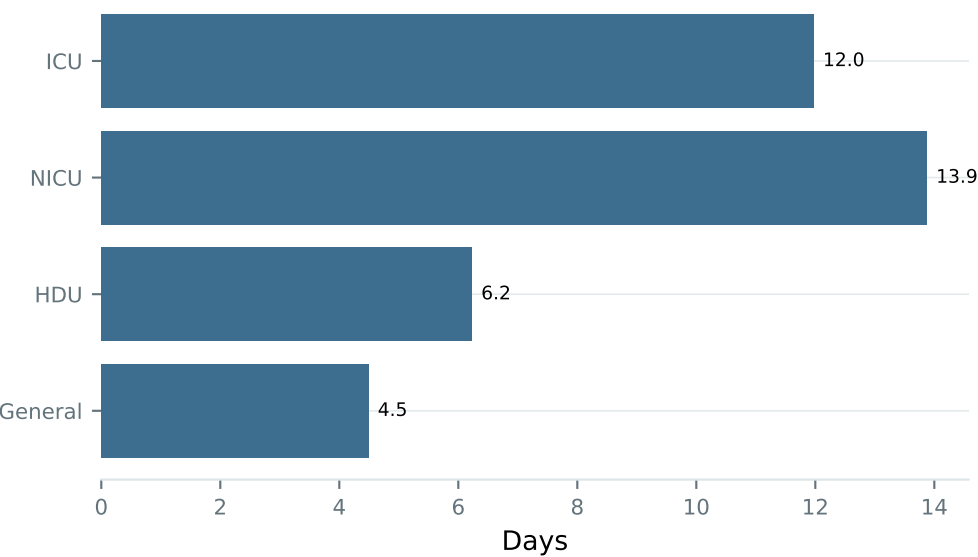
Monthly admissions | Latest 24 months



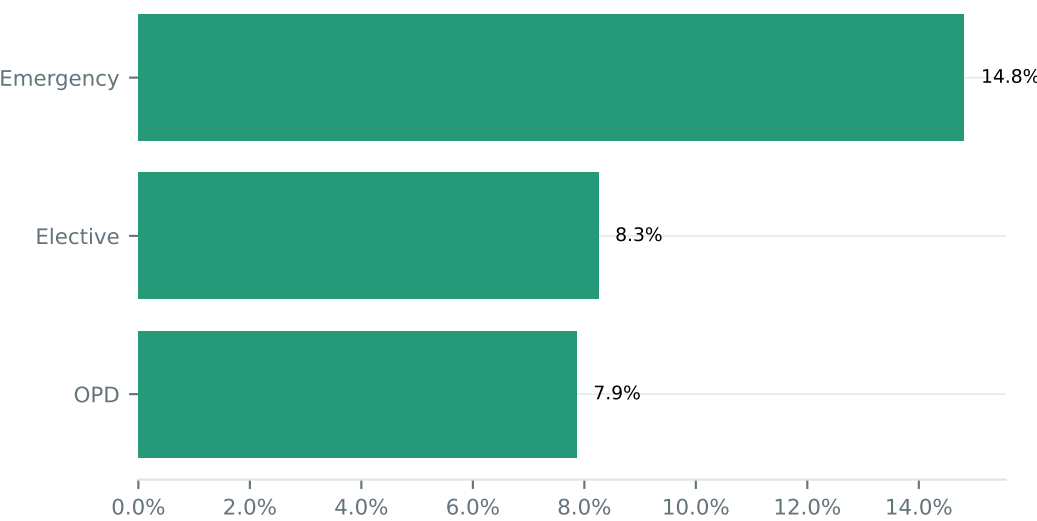
30-day readmission by observed ward



Average LOS by observed ward



Readmission by observed admission type



Illustrative readmission threshold is not hospital-approved. High-complexity score ≥ 10 : 34,081 encounters, 21.3% readmission.

Financial and Claims Performance

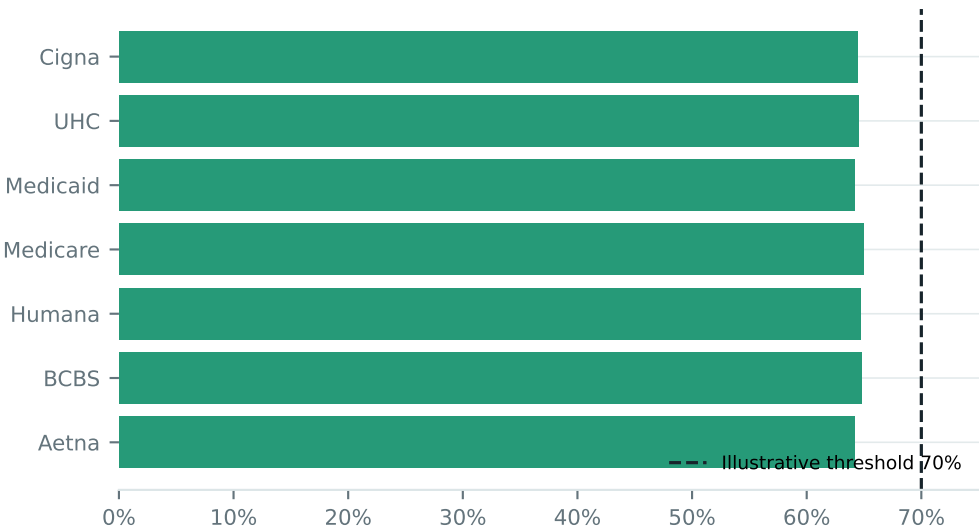
Observed aggregate claims evidence | No audited admission-level attribution

Period: 03 Jan 2015 - 10 May 2025 | Generated: 31 Jul 2026 06:58 UTC

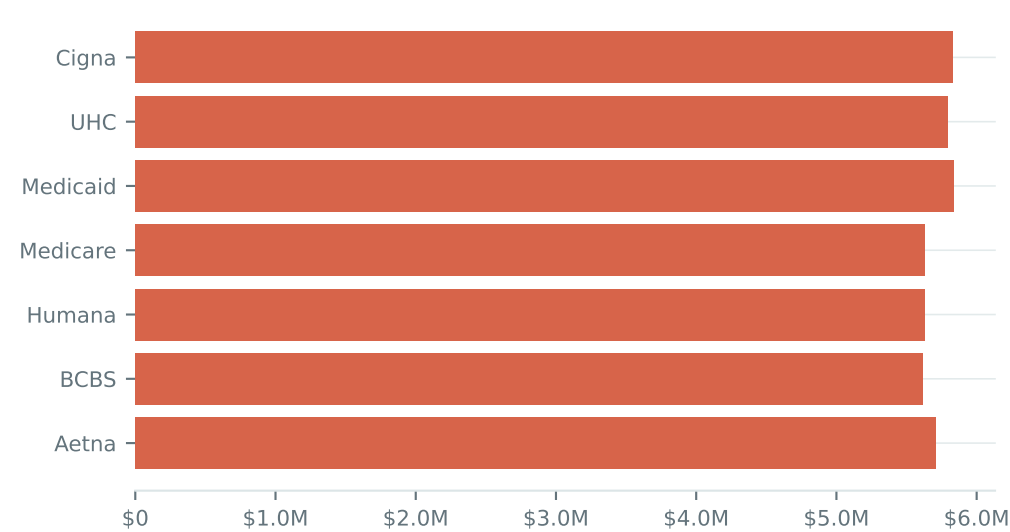
Monthly billed versus paid | Latest 24 months



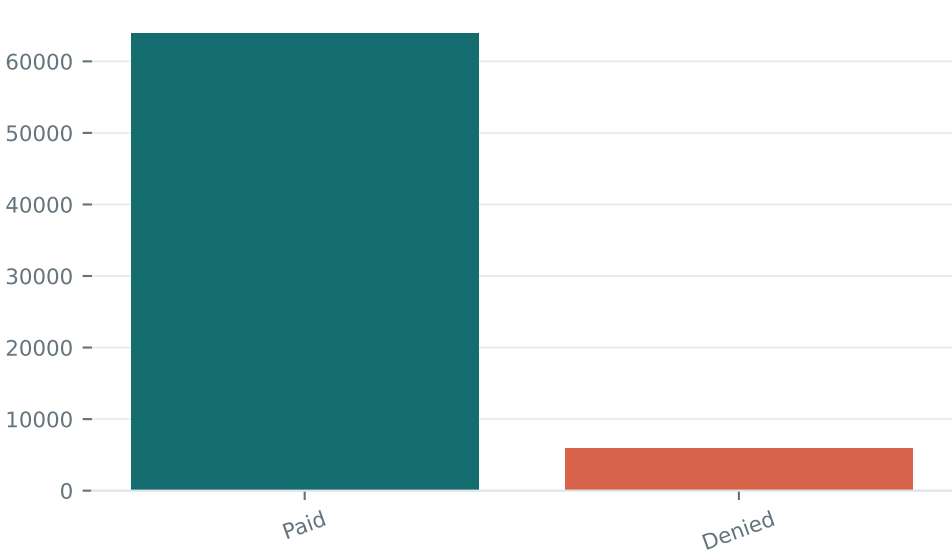
Collection rate by observed payer



Billed-to-paid gap by payer



Observed claim status distribution



Claims: 70,000 rows versus 120,000 admissions. Overall approval 91.4%; collection 64.5%. A surrogate link is used only for portfolio workflow demonstration.

Decision Governance and Delivery Plan

Core analytical evidence is separated from sandbox demonstrations

Period: 01 Jan 2015 - 30 Dec 2024 | Generated: 31 Jul 2026 06:58 UTC

MODEL READINESS

Model	Portfolio tier	Holdout result	Target provenance	Decision use
Readmission	Core analytical	ROC AUC 0.736	Observed	Screening research only
Bed occupancy	Core analytical	MAE 6.38 beds	Observed census + capacity assumption	Capacity scenario
Waiting time	Sandbox	RMSE 0.074 min	Simulated formula	Pipeline demonstration
Revenue	Sandbox	R2 0.144	Surrogate + simulated	Scenario demonstration

ACCOUNTABLE ACTION PLAN

Priority	Action	Owner	Timeframe	Success measure
P1	Prioritize ICU discharge review and post-discharge follow-up.	Clinical Quality Lead	30 days	Root-cause review completed and approved ward target set
P1	Review payer-specific billed-to-paid gaps and aged claims.	Revenue Cycle Manager	30 days	Collection rate monitored against illustrative 70% threshold
P2	Capture arrival, triage, and service-start timestamps before staffing optimization.	Operations Data Owner	60 days	Observed queue timestamps pass completeness checks
P2	Replace surrogate claim links and derived master data	Data Governance Lead	90 days	Governed encounter, provider, department, and patient keys

DERIVED DIMENSIONS

Doctor, department, demographics, medicine, appointment, waiting-time, and admission-claim relationships are deterministic portfolio constructs. Observed hospital, ward, dates, clinical fields, readmission, payer, and aggregate billing evidence lead all executive conclusions.

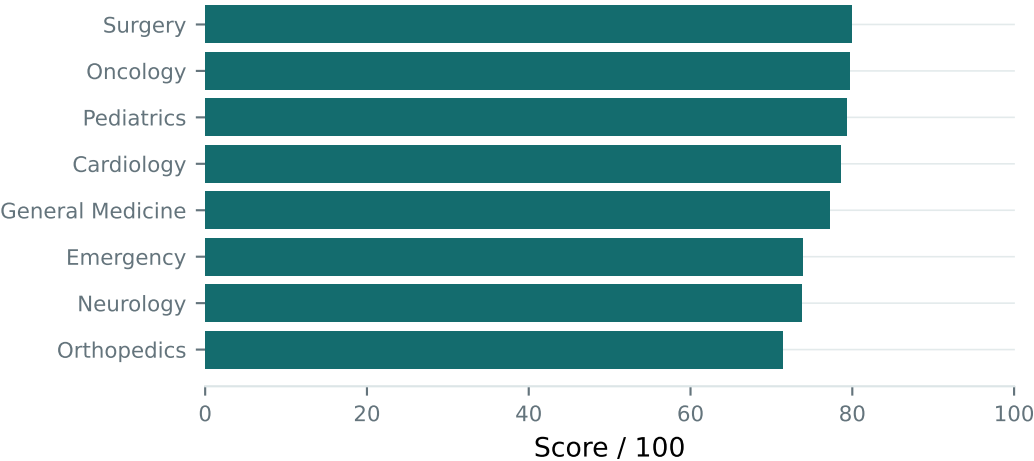
Hospital Command Center

Latest observed snapshot 30 Dec 2024 | Forecasts and assumptions are explicitly labelled

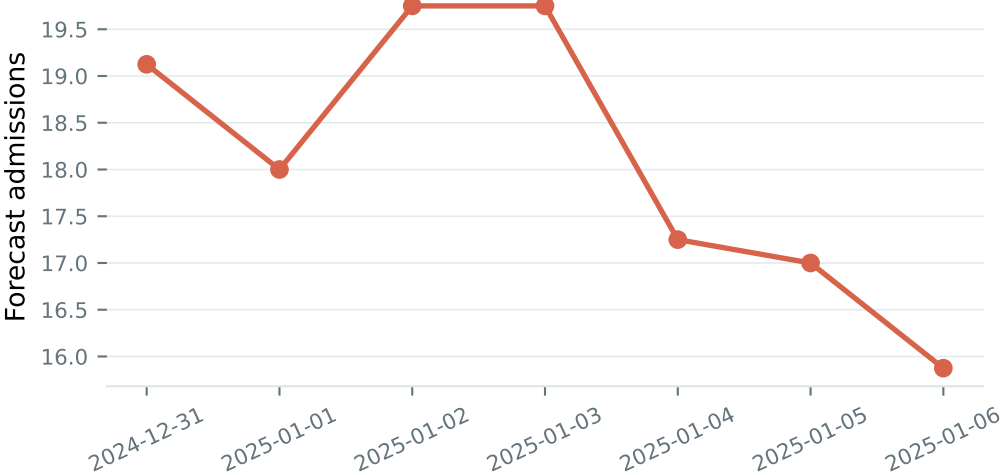
Period: 01 Jan 2015 - 30 Dec 2024 | Generated: 31 Jul 2026 06:58 UTC

PATIENTS TODAY	BED OCCUPANCY	EMERGENCY WAIT	CRITICAL PATIENTS	DOCTOR UTILIZATION	EFFICIENCY SCORE
36	52.6%	50 min*	14	75%*	76/100*

Department efficiency score*



Next-week emergency outlook | 127 patients



ACCOUNTABLE RECOMMENDATIONS

Priority	Signal	Action	Owner	Timeframe
P1	11.8% hospital-wide readmission	Prioritize ICU discharge review and post-discharge follow-up.	Clinical Quality Lead	30 days
P1	64.5% of billed value collected	Review payer-specific billed-to-paid gaps and aged claims.	Revenue Cycle Manager	30 days
P2	50 minute simulated emergency wait	Capture arrival, triage, and service-start timestamps before staffing optimization.	Operations Data Owner	60 days

* Doctor utilization and department are derived; emergency wait is simulated; capacity uses the configured 500-bed assumption. Peak-hour forecasting is withheld because arrival timestamps are unavailable.